ASSIGNMENT-1

1) What are the characteristics and classification of computer ?

Ans. The computer system can be classified on the following basis :-

→ Classification On the basis of Size :-

1) SUPER COMPUTER → The super computer are the most high performing system. A supercomputer is a computer with a high level of performance compared to the general purpose computers.

2) MAINFRAME COMPUTER → These are commonly called as big iron, they are usually used by big organisation for bulk data processing, transaction processing and are widly used as the servers as these systems has a higher processing capability as compared to the other classes of computers.

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3). MINI COMPUTER → These computers came into the market in mid 1960s and were sold at a much cheaper price than the main frames, they were actually designed for control, instrumentation, human interaction, and communication switching as distinct from calculation and record keeping, later they became very popular for personal uses with evolution.

4). MICRO COMPUTER → A microcomputer is small, relatively inexpensive computer with a micro processor as its CPU. It includes a microprocessor, memory, and minimal I/O circuitry mounted on a single printed circuit board. The forerunners of these computers, mainframes and minicomputer were comparatively much larger, hard to maintain and more expensive. They actually formed the foundation for present day microcomputers and smart gadgets that we use in day to day life.

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→ Classification on the basis of functionality :-

1). SERVER → Servers are nothing but dedicated computer which are set-up to offer some services to the clients. They are named depending on the type of service they offered.
Eg: Security servers, database servers.

2). WORKSTATION → Those are the computers designed to primarily to be used by single users at a time. They run multi-user operating system.

3). INFORMATION APPLIANCES → They are the portable device which are designed to perform a limited set of tasks like basic calculations playing multimedia, browsing internet etc.

4). EMBEDDED COMPUTER → They are the computing devices which are used in other machine to serve limited set of requirement. They follow instructions from the non-volatile memory and they are not require to execute reboots or reset.

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→ Classification on the basis of data handling :-

1). ANALOG → An analog computer is a form of computer that uses the continuously changeable aspects of physical facts such as electrical, mechanical or hydraulic quantities to model the problem being solved.

2). DIGITAL → A computer that perform calculation and logical operations with quantities represented as digit, usually in the binary number system of '0' and '1', computer capable of solving problems by processing information expressed in discrete form

3). HYBRID → A computer that processes both analog and digital data, hybrid computer is a digital computer that accepts analog signals converts them to digital and processes them in digital form.

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→ Characteristics of Computer System :-

1). SPEED → Executing mathematical calculation, a computer work faster and more accurately than human.

2) DILIGENCE → A human cannot work for several hours without resting, Yet computer never tires.

3). RELIABILITY → A computer is reliable. The output results never differ unless the input varies.

4). VERSATILITY → Versatility refers to a capacity of computer. Computer perform different types of task with the same accuracy and efficiency.

5) MEMORY → A computer can store millions of records these records may be accessed with complete precision.

6). ACCURACY → When a computer perform a computation or operation, the chance of errors occuring are low.

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2) What is generation in computers ? Explain

2). The development of computers has gone through different generations, each generation marked by significant advancement in term of technology and architecture. These generations are classified as follow :-

1). FIRST GENERATION :-

→ The timeline for the first generation computer was 1940 to 1956. (1946-1954)

→ The 1st gen. computers were developed using vacuum tubes or thermionic value machine.

→ Punched cards and paper tape were used as input / output.

→ These were bigger in size and more expensive

→ These computers were worked on binary coded concepts (0-1)

→ Ex → ENIAC, EDVAC etc.

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2). SECOND GENERATION :-

→ The timeline for the second generation computers was 1956 to 1963.

→ Transistors were used to develop

→ Punched cards & magnetic tapes were used for I/O.

→ Magnetic core memory was used

→ Ex = UNIVAC , IBM 1401 , IBM 7090

3). THIRD GENERATION :-

→ The timeline for the third generation computers was 1963 to 1971

→ Integrated circuit (IC) was used to develop

→ Small in size as compared to 1st & 2nd generation

→ Magnetic tape , keyboard , monitor , printer devices were used as input and output

→ Ex = UNIVAC , IBM 360 , IBM 370

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4. FOURTH GENERATION :-

→ The timeline for the fourth generation computer was 1972 to 2010.

→ Microprocessor technology was used to develop.

→ Very less power consuming and affordable.

→ Key board, pointing device, optical scanning, monitor, printer devices were used for I/O.

→ Ex = IBM PC, STAR 1000, Apple.

5). FIFTH GENERATION :-

→ The timeline for the fifth generation computer is from 2010 to till date.

→ These computers are based on artificial intelligence, Ultra large-scale integration, quantum computation nanotechnology, parallel processing technology.

→ Very fast and multiple task could be performed simultaneously.

→ Ex- laptops, tablets, smartphones etc.

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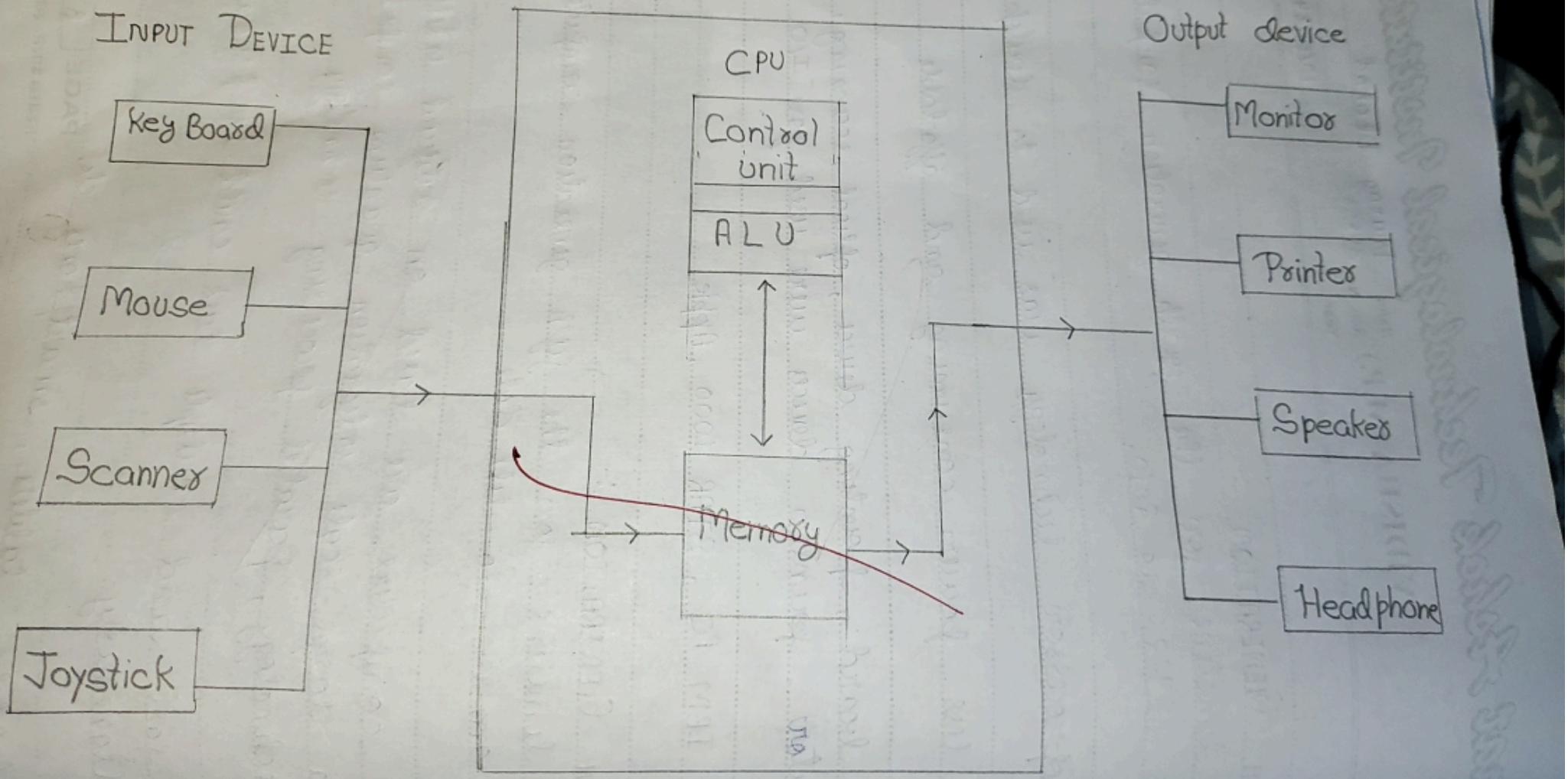
BASIC Architecture of a Computer



3). What is the basic structure of computers in computer organisation and architecture?

3). Three components from the basic structure and function. These three components are input devices, CPU, & output device. CPU can also divided into two parts that form the basic structure of computers. These two parts are ALU and CU.

a). INPUT UNIT :-

Input units are the components or devices of the computer by which we can enter any data into the computer. Input devices are responsible for translating all the information that we add to the computer into a form only understand by the computer. All computer types have an input unit.

b). THE CENTRAL PROCESSING UNIT (CPU) :-

The central processing or CPU is known as the computer's brain. The CPU play an important role in forming the

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basic structure of computers, CPU it does all the data operation. The CPU is responsible for storing data, intermediating results and instructions of programs. The central processing unit controls the operations of every part of the computers

A CPU has three components. These three components are :-

1). Control unit
2). The memory unit
3). ALU / Axithmetic logic unit.

b(i) MEMORY UNIT :-

Another important component of all computers types is the memory unit. A memory unit is a place where all the data such as programs as well as ~~quick for a CPU~~ files, are stored. It become easy as well as quick for a CPU to access all the information through the memory unit

Two types of memory units are main :-

1). RAM (Random Access Memory)
2) ROM (Read Only Memory)

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RAM is responsible for storing data when used, whereas ROM is responsible for retaining data when power is not provided.

The type of memory and the size of memory determines the performance of the system of computer.

b(ii) CONTROL UNIT :-

Every body knows of the CPU, but there is one more components that affects the processing of data. This unit is known as the control unit (CU). The control unit does not receive any form of input from users; rather it is responsible for sending commands to other computer components, for example the instruction decoder and register. The computer's control unit highly affects everything that happens inside the computer.

It handles everythings from the execution of instruction to the running of programs. The control unit does not have any physical form, but still, it form the

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the basic structure of computer.

9. OUTPUT UNIT :-

The device that helps us get all the required information from the computer is known as the output unit. The output unit act as a linkage between users as well as computer.

Output devices are responsible for translating the computer's output into a form that is understandable by the users.

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4. Discuss about instruction set, memory and I/O devices.

4). INSTRUCTION SET :-

An instruction is a set of codes that the computer processors can understand. The code is usually in 1s and 0s, machine language. It contains instructions or tasks that control the movement of bits and bytes within the processor.

Example of some instruction sets -

1). ADD - add two no. together
2) JUMP - jump to designated RAM address
3) LOAD - Load information from RAM to the CPU.

→ Types of instruction Set :-

a). Reduced Instruction Set Computer (RISC) :-

A number of computers designers recommended that computers use fewers instruction with simple constructs so that they can be executed much faster within the CPU without having to use memory as often.

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This type of computer is called a reduced instruction set computer.

→ Characteristics of RISC :-

1). Relatively few instructions.
2). Relatively few addressing modes.
3). Memory access limited to load and store instructions.
4). Single - cycle instruction execution.
5). Fixed length, easily decoded instruction format.
6). All operation done within the resistors of the CPU.

b). Complex Instruction Set Computer (CISC) :-

CISC is a computer where single instruction can perform numerous low - level operations like a load from memory and a store from memory, etc. The CISC attempts to minimize the no of instructions per program but at the cost of an increase in the number of cycle per instruction.

The design of an instruction set for a computer must take into consideration

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not only machine language constructs but also requirement imposed on the use of high level programming language.

The goal of CISC is to attempt to provide a single machine instruction for each statement that is written in a high level language.

→ Characteristics of CISC :-

1) A large number of instruction typically from 100 to 250 instruction.
2) Some instruction that perform specialized task and are used infrequently.
3) Variable length instruction formats
4) Instruction that manipulate operands in memory
5) A large variety of addressing modes typically from 5 to 20 different modes.
6) Instruction may take more than a single cycle to get executed

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• COMPUTER MEMORY :-

Computer memory is just like the human brain. It is used to store data / information and instruction. It is a data storage unit or a data storage device where data is to be processed and instructions required for processing are stored. It can store both the input and output can be stored here.

→ TYPES of MEMORY :-

a). Primary Memory → It is also known as the main memory of the computer system. It is used to store data and programs or instruction during computer operations.

b). Secondary Memory → It is also as auxillary memory and backup memory. It is a non-volatile memory and used to store a large amount of data.

c). Cache Memory → It is a type of high-speed semiconductor memory that can help the cpu run faster.

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• INPUT & OUTPUT DEVICES :-

a) INPUT DEVICE → Input devices are the devices that are used to send signals to the computer for performing tasks. The receivers at the end is the CPU, which work to send signals to the output devices.

Some of the classification of input devices are :-

1) Keyboard
2) Painting
3) Game controller
4) Visual etc.

b) OUTPUT DEVICE :-

Output devices are the devices that show us the result after giving the input to a computer system. Output can be of many different form like image, graphic audio, video etc.

Some of the output devices are :-

1) Monitor
2) Television
3) Printers
4) Speakers
5) Projectors
6) Video card
7) etc.

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5). Difference between System software & application software.

5) SYSTEM	APPLICATION
1) System software is meant to manage the system resources. It serve as platform to run	1) Application software help perform a specific set of function for which they have been designed
2) Low language is used	2) High language is used
3). Work in the background	3) Works in the foreground
4). Interact with the hardware directly	4). Does not take the hardware into consideration as it does not interact directly
5) Ex :- Operating system, programming language, communication software etc	5) Ex→ word doc, spreadsheet, database, internet explorer.

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ASSIGNMENT → 2

1). What is the difference between data type & Variables.

1). DATA TYPE :-

• A data type defines the kind of value a variable can store. It specifies the nature of the data, such as whether it is a number, a string, a boolean etc.

• Data type determine how much space the data will occupy in memory and what kind of operations can be performed on that data

• Examples of common data type include :

1). Integers (int) : Whole numbers (e.g., 5, -3, 100)

2). Float (float) : Decimal numbers (e.g., 3.14, -0.001)

3). Boolean (bool) : Represent true or false value

→ VARIABLE :-

• A variable is a symbolic name given to a memory location that stores a value. A variable hold data of a specific type.

• Variable are used to store data temporarily and can be manipulated throughout the program.

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• Variable are associated with a data type, which dictates what kind of value they can store.

• Variable can change their value during the program execution (they are mutable), unlike constants that remain fixed.

• For Example :-

```
int age = 25;  // 'age' is a variable of type
int that stores the value 25
```

→ KEY DIFFERENCE :-

1). Data type define the kind of data a variable can store, whereas a Variable is the container that holds the actual data.

2). Data type specify the memory size and operations allowed on the data, while variable are used to store and reference the data of a particular data type.

3). In C, a variable must be declared with a specific data type before it is used to hold data.

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2). Explain various types of data types that are found in the C programming language?

2). 1). PRIMITIVE DATA TYPES :-
Primitive data types are the most basic data types that are used for representing simple values such as integers, float, characters etc.

i). Integer Types (int) :- Used to store whole numbers (both +ve and -ve) without decimals.
→ Size = 4 bytes, Format Specifier = %d, Range → -2147483648 to 2147483647
→ Syntax = int var_name ;

ii). Character Data Type → character data type allows its variable to store only a single character.
→ Range → (-128 to 127) or (0 to 255), Size = 1 byte
→ Format specifier = %c, Syntax = Char var_name ;

iii). Float data Type → Float in C is used to store decimal numbers with single precision.
→ Range → (1.2E-38 to 3.4E+38), size = 4 bytes
→ Format specifier = %f, Syntax = float var_name ;

iv). Double Data Type → It store decimal no. with double precision. It is used to define numeric values which hold numbers with decimal value in C.

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→ Range = 1.7E-308 to 1.7E+308, Size = 8 bytes, Format Specifier = %lf
Syntax = double var_name

v). Void Data Type → The void data type in C is used to specify that no value is present. It does not provide a result value to its caller.

2). DERIVED DATA TYPE → The data types that are derived from the primitive or built in datatype are referred to as derived data types

Data type → array, pointers, function

3). USER DEFINED DATA TYPE → The user defined data types are defined by the users himself.

Data type → structure, union, enum.

3). Why do we need modifiers in the C programming language.

3). In C language, modifiers are used to modify the default behaviour of data types, helping to define the size, range, or representation of the d variables. They provide more control over how data is stored and accessed in

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memory. Here are some key reasons we need modifiers in C:

1). Make data types more specific → modifiers help make data type more specific, similar to how saying "dark chocolate bar" is more specific than just "chocolate bar".

2). ~~Ay~~ Adjust memory allocation → Modifiers indicate how much memory to allocate to a variable

3). Modify data type range → modifiers can increase or decrease the range of data types.

Some examples of modifiers in C include:

1). Short → Limits the user to store small integer values from -32768 to 32767

2). Long → Allow the user to store very large number

3). Signed → The default modifier for int and char data types, allow the user to store -ve & +ve value

4). Unsigned → Allow the user to store only +ve value in given data type

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4. Describe various types of operators used in C programming language and also explain the operators that are used in structure with example.

4). In C programming, operators are special symbols that perform operations on variables and values. Operators can be classified into several categories based on the type of operation they perform.

→ TYPES of OPERATIONS IN C :-

1). ARITHMETIC OPERATION → These operators are used to perform basic arithmetic operations.

+ : Addition , - : Subtraction , * : Multiplication

/ : Division , % : Modulus

Ex→ int sum = a+b ; // sum = 15 { a = 10 , b = 5 }

2). RELATIONAL OPERATORS → These operators are used to compare two values.

== : Equal to , != : Not equal to , > : Greater than

<= : less than or equal to

Ex → int a = 10 , b = 5 ;
if (a > b) {
// This condition is true
}

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3). LOGICAL OPERATORS → Logical operators are used to combine conditional expressions.
&& : Logical AND , || : Logical OR
! : Logical NOT

Ex:- int a=10 , b=5
if (a>b && b>0) { // This condition is true }

4). BITWISE OPERATORS → These operators are used to manipulate individual bits of an integers.
& : Bitwise AND , | : Bitwise OR , ^ : Bitwise XOR
~ : Bitwise NOT , << : Bitwise left shift , >> : Right shift

Ex → int result = a & b ; // result = 0001 {a= 0101, b= 0011}

5). ASSIGNMENT OPERATORS → These operators are used to assign values to variables.
= : Simple Assignment , += : Add & Assign , -= : Subt. & Assign
/= : Divide & Assign , *= : Multiply & Assign , %= : Modulus & assign

Ex → int a = 10 ;
a += 5 ; // a = a+5 ⇒ a = 15

6). Increment & Decrement Operators → These are used to increase or decrease the value of a variable by 1.

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++ : Increment , -- : Decrement

Ex → a++ ; // a become 6 { Int a = 5 }
--a ; // a become 4

7). Conditional or Ternary Operation → It is used as a shorthand for an if-else statement.
? : : Conditional operator

→ OPERATORS USED WITH STRUCTURE IN C :-

1). Dot Operator (.) → The dot operator is used to access members of a structure through a structure variable.

2). Arrow Operator (->) → The arrow operator is used to access members of a structure when you have a pointer to a structure.

5). Explain type casting and type conversion in C ?

5). In C programming, type casting and type conversion are both techniques used to change the data type of a variable, but they are used in different contexts and in slightly different ways.

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1. TYPE CONVERSION (IMPLICIT CONVERSION)

Type conversion, also known as implicit conversion, is the automatic conversion of one data type to another by the compiler. It happens when you perform an operation involving two different type and C automatically promotes or demotes the data types to match the expected type.

Ex → int a = 5
float b = 4.2 ;
float result = a + b ; // 'a' is implicitly convert to float.

2). TYPE CASTING (EXPLICIT CONVERSION)

Type casting is when you explicitly convert one data type into another using a cast operator. This is necessary when you need to convert a value from one type to another, and you want to control how the conversion happens.

→ Type casting is done by replacing the desired data type in parentheses before the value you want to cast.

Ex → float a = 5.75 ;
int b = (int) a; // Explicitly cast 'a' to an integer.

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→ Types of Type Casting :-
1). Implicit type casting
2). Explict type casting

6). Discuss priority and associativity of operators in C.

6). In C programming, priority and associativity are important concepts when dealing with expressions involving multiple operators. They determine the order in which operators are evaluated and how expressions are grouped.

1). OPERATOR PRIORITY → It define the oder in which operators are evaluated in an expression. Operators with higher precedence are evaluated before those with lower precedence.

For Example → In the expression a+b*c, the multiplication (*) ~~has higher~~ is performed first because multiplication has higher precedence than addition (+)
20 + 10 * 10 = 20 + 100 = 120

2). OPERATOR ASSOCIATIVITY → Associativity define the direction in which operators of the same precedence level are evaluated.

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ii). Left - to - right associativity → Most operators in C have this associativity, meaning they are evaluated from left to right.

Ex → In the expression a - b - c (i.e (a-b)-c)

ii). Right - to - left associativity → Some operators, such as assignment (=), conditional (?:), and increment / decrement operators (++, --) are evaluated right to left.

7). Difference between argument and parameter.

7). ASPECT	PARAMETER	ARGUMENT
• Defination	Variables in the function definition, used to receive values	Actual value or expression passed to the function when called
• Location	Declared in the function signature	Giving during function call.
• Role	Acts as a placeholder for values to be passed	Supplies the value to the function.

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• Example	int a, int b in void add (int a, int b)	5 and 3 in add (5,3)

→ Parameter Passing → Refers to the process of providing values (arguments) to a function when it is called, so that the function can operate on those values. In C, there are several ways to pass parameters to a function, and the choice of method affects how the values are handled within the function. The main techniques for parameters passing are :-

1) Call by Value
2) Call by reference

• Call by Value → Passes a copy to the value to the function. Changes do not affect the original value.

Ex → void addTen (int num)

• Call by Reference → Passes the address of the variable, allowing the function to modify the original value

Ex → void addTen (int * num)

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8). What is recursion ? Explain different types of recursion.

8). Recursion is a programming technique where a function calls itself to solve a problem. A recursive function typically has two main parts
1). Base Case → A condition that stops the recursion and prevents infinite loops.

2). Recursive Case → The part where the function calls itself with modified arguments to break down the problem into smaller subproblems.

→ Types of Recursion :-
1) Direct recursion → In direct recursion, a function calls itself directly in its body.

2). Indirect recursion → In indirect recursion, a function calls another function, and that function calls the original function, forming a cycle.

3). Tail recursion → Tail recursion occurs when the recursive call is the last operation in the function. It can be optimized by compilers to avoid creating new stack frames.

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4) Head Recursion → Head recursion is when the function makes a recursive call before performing any other operations. In other words, the recursive call happens at the beginning of the function.

5). Tree Recursion → Tree recursion happens when a function makes multiple recursive calls within the same function. This results in a branching structure, like a tree.

9). Define structure ? Show how union is different from structure through an example

9) A structure is a user defined data type in C that allows you to group different data types together under a single name. Each element of a structure is called a member, and each member can have different data types. Structures are particularly useful when you want to represent an entity with multiple attributes.

→ Key difference :

i) Memory Allocation

• Structure → Each member of a structure gets its own memory location. The total size of the structure is the sum of the sizes of

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its member

• Union → All members share the same memory location. The total size of the union is the size of the largest member.

ii) Accessing Data :

• Structure → All member can hold different values simultaneously.

• Union → Only one member can hold a value at a time. The last assigned value overwrites previous one.

10). Explain file handling and its function.

10) File handling is the process of storing, updating, opening, reading, writing, and deleting files in a program. It allows users to create data file such as text or binary files to store input or output from a program for future use.

→ File handling has several benefits, including :

1) Data reusability → data can be reused across different systems.

2). Large dataset manipulation → large datasets can be efficiently manipulated.

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3). Time & effort saving → file handling can save time and effort, and reduce the risk of flawed coding.

→ Some functions used in file handling in C include :
1). fopen() → used to open file with different modes, such as read, write
2). fwrite() → used to read and write data in binary format.
3). fread() → used to read and write data in binary format
4). fgetc() → used to read single character
5). fputc() → used to write a single character
6). fgets() → Used to read a line of text.
7). fputs() → Used to write a string.
8). fprintf() → used to write formatted text
9). fscanf() → used to read formatted text
10). rewind() → used to reset the file position indicator.

11) Explain :-
a) Various types of computer security.
a) Here are some types of computer security :-

1). Network Security :-
Protects a network's integrity and data from threats like cyber attacks and intrusions.

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It involves a variety of hardware and software tools, including firewalls, antivirus software.

2). Application security → protects software from threats throughout its lifecycle.

3). Cloud security → Protects data and systems hosted on cloud platforms such as Amazon Web services (AWS) and google cloud.

4). Digital security control → Includes usernames and passwords, two-factor authentication, antivirus software and firewalls.

5). Cybersecurity controls → Includes anything designed to prevent attacks on data, including DDoS mitigation and intrusion prevention systems.

b). Principles of operating System :-

i) Intermediary → The operating system acts as an intermediary between the users and the computers hardware.

ii). Resource management → The operating system manages resources like memory, CPU & I/O device

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iii). Multitasking → The OS allows multiple applications to run at the same time.

iv). Security & protection → The operating system implements security measures to protect system resources data, and user privacy.

v). User interface → The OS provides a user-friendly interface for interacting with the computer system

vi). Process, file system and device management.

c). Difference b/w machine learning and AI :-

c). Aspects	Artificial Intelligence (AI)	Machine Learning (ML)
• Defination	Broad field focusing on creating intelligent system that mimic human behaviors	A subset of AI that focuses on learning from data and improving over time
• Scope	Includes various technologies like reasoning, problem	Focuses on algorithms and data-driven learning

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	solving and learning	
• Goal	To build machine that can simulate human intelligence	To enable systems to learn and make decisions based on data
• Approach	Can involve rule-based systems or learning based system	Primarily based on data and algorithms to identify pattern
• Learning	May involve logic, reasoning as ML	Always involves learning from data
• Example	A self-driving car making autonomous decision in traffic	A spam filter that classifies emails based on historical data

12). Define :-

a). Data Science :-

Data science is the scientific study of data to extract knowledge and insights for making informed decisions and predictions

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Data science is a multidisciplinary field that combines principles and practices from mathematics, statistics, business, AI and computer engineering.

Data scientists use tools like data analytics, machine learning, and ~~or~~ AI to :

→ Collect, Analyze and interpret large amount of data
→ Uncover hidden pattern and trends.

b). Cloud Computing :-

Cloud computing refers to the delivery of computing services such as storage, processing power, database networking, software and analytics - over the internet (the cloud), instead of relying on local servers or personal device. This model allows businesses and individuals to access and use technology resources on demand, without the need for owning or managing physical infrastructure.

→ Benefit of cloud computing

1). Cost efficiency
2) Scalability & flexibility
3) Accessibility & mobility

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4). Security & reliability
5). Automatic updates & maintenance

→ Common use of cloud computing :-
1). Data storage & backup
2). Web hosting
3). Big data and analytics
4). Collaboration tools

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